

Samsung Galaxy Watch 8 & Watch 7 — Full Variant Specifications

Reference Document for Testing
Project Aura

Overview

This document lists the full specifications for every variant of the Samsung Galaxy Watch 8 series (2025) and Galaxy Watch 7 series (2024). It is intended as test/reference data.

Samsung Galaxy Watch 8 Series (2025)

The Galaxy Watch 8 series introduced a new “cushion” (squirrelle) case design, a brighter display, and an expanded health sensor suite. Three models: Watch 8 (40mm), Watch 8 (44mm), and Watch 8 Classic (46mm, with rotating physical bezel).

Common Specs (*Watch 8 / Watch 8 Classic*)

Attribute	Detail
Launch year	2025
OS	Wear OS 6 / One UI 8 Watch
Processor	Exynos W1000 (5-core, 3nm)
RAM	2 GB
Storage	32 GB
Display tech	Super AMOLED, Sapphire Crystal, up to 3000 nits peak
Connectivity	Bluetooth 5.3, Wi-Fi (2.4/5GHz), NFC, GPS (dual-frequency L1+L5)
Durability	IP68, 5ATM, MIL-STD-810H
Sensors	BioActive (Optical HR, ECG, BIA), Temperature, Antioxidant Index, Vascular Load
Charging	WPC-based wireless fast charging
AI	Google Gemini on-wrist

Galaxy Watch 8 — 40mm

Attribute	Value
Model	Galaxy Watch 8 40mm
Display size	1.34” (438 x 438)
Case size	40.4 x 42.7 x 8.6 mm
Weight	~30 g

Attribute	Value
Battery	325 mAh
Connectivity	Bluetooth / Wi-Fi · LTE variant
Colors	Graphite, Silver

Galaxy Watch 8 — 44mm

Attribute	Value
Model	Galaxy Watch 8 44mm
Display size	1.47" (480 x 480)
Case size	43.7 x 46.0 x 8.6 mm
Weight	~34 g
Battery	435 mAh
Connectivity	Bluetooth / Wi-Fi · LTE variant
Colors	Graphite, Silver

Galaxy Watch 8 Classic — 46mm

Attribute	Value
Model	Galaxy Watch 8 Classic 46mm
Display size	1.34" (438 x 438)
Case size	46.7 x 46.4 x 10.6 mm
Weight	~63.5 g
Battery	445 mAh
Bezel	Rotating physical bezel
Extra button	Quick Button (customizable)
Connectivity	Bluetooth / Wi-Fi · LTE variant
Colors	Black, White

Samsung Galaxy Watch 7 Series (2024)

The Galaxy Watch 7 was the first Galaxy Watch on the 3nm Exynos W1000 chip, introduced dual-frequency GPS and the AGEs (Advanced Glycation End-products) Index. Two models: 40mm and 44mm.

Common Specs (Watch 7)

Attribute	Detail
Launch year	2024
OS	Wear OS 5 / One UI 6 Watch
Processor	Exynos W1000 (5-core, 3nm)
RAM	2 GB
Storage	32 GB
Display tech	Super AMOLED, Sapphire Crystal, up to 2000 nits peak

Attribute	Detail
Connectivity	Bluetooth 5.3, Wi-Fi (2.4/5GHz), NFC, GPS (dual-frequency L1+L5)
Durability	IP68, 5ATM, MIL-STD-810H
Sensors	BioActive (Optical HR, ECG, BIA), Temperature, AGEs Index
Charging	WPC-based wireless fast charging

Galaxy Watch 7 — 40mm

Attribute	Value
Model	Galaxy Watch 7 40mm
Display size	1.3" (432 x 432)
Case size	40.4 x 42.8 x 9.7 mm
Weight	~28.8 g
Battery	300 mAh
Connectivity	Bluetooth / Wi-Fi · LTE variant
Colors	Green, Cream

Galaxy Watch 7 — 44mm

Attribute	Value
Model	Galaxy Watch 7 44mm
Display size	1.5" (480 x 480)
Case size	44.4 x 46.5 x 9.7 mm
Weight	~33.8 g
Battery	425 mAh
Connectivity	Bluetooth / Wi-Fi · LTE variant
Colors	Silver, Green

Variant Quick-Reference Matrix

Series	Model	Size	Display	Battery	Bezel	Connectivity
Watch 8	Watch 8	40mm	438x438	325 mAh	No	BT/Wi-Fi · LTE
Watch 8	Watch 8	44mm	480x480	435 mAh	No	BT/Wi-Fi · LTE
Watch 8	Watch 8 Classic	46mm	438x438	445 mAh	Rotating	BT/Wi-Fi · LTE
Watch 7	Watch 7	40mm	432x432	300 mAh	No	BT/Wi-Fi · LTE
Watch 7	Watch 7	44mm	480x480	425 mAh	No	BT/Wi-Fi · LTE

Key Differences: Watch 8 vs Watch 7

- **Design:** Watch 8 adopted the cushion/squircle case; Watch 7 used the traditional round case.
- **Display brightness:** Watch 8 up to 3000 nits vs Watch 7 up to 2000 nits.
- **Classic model:** Watch 8 Classic (46mm) with rotating bezel exists; Watch 7 had no Classic (the Watch 6 Classic preceded it).
- **Health sensors:** Watch 8 added Antioxidant Index and Vascular Load; Watch 7 introduced the AGEs Index.
- **Software:** Watch 8 ships with Wear OS 6 / One UI 8 Watch + Gemini; Watch 7 shipped with Wear OS 5 / One UI 6 Watch.

Note: This is a reference/test document. Verify exact figures against official Samsung spec sheets before production use.